

Sound¹

Disha Kuzhively

March 10, 2026

¹ Draft version as of March 10, 2026

This article was prepared by the author to serve as a draft for TNSCERT class 8 chapter on the concept of Sound for TNSCERT during the the Tamil Nadu Curriculum Framing(TNCF) workshop that was held in Asha Nivas, Chennai from 13-11-2025 to 15-11-2025, and 09-03-2026 to 10-03-2026.

The right side margin of each page contains notes for teachers, and prompts to activities that the teacher can initiate in the classroom.

Every day we experience many natural and human-made sounds around us. For example: a school bell ringing, raindrops falling on the ground, a dog barking, or a phone ringing.

EXERCISE: Write down all the sounds that you hear in your daily life.

Where do these sounds come from? How are these sounds produced? In this chapter, we will explore how sound is produced and how it propagates² from one place to another. We will also learn that sound can be understood as a wave.

EXERCISE: Close your eyes for one minute and listen carefully. Write down all the sounds you hear.

² Here the word 'propagate' is used instead of simpler and familiar word like 'travel' to avoid misconceptions that students create, e.g. The word travel suggests that 'sound' is a moving object. Words are chosen to ensure that students develop a scientifically accurate mental model of the concept.

How is Sound Produced?

Activity: Vibrating Scale

Materials required:

1. Scale
2. Desk or table

Place the scale on the desk so that about one-third of it is over the edge. Hold the part on the desk with one hand. Push the free end of the scale downward. Release it quickly.

OBSERVE: Do you hear a sound? Do you see the scale move up and down rapidly? What type of motion is this?

Activity: Rubber Band Instrument

Materials required:

1. Rubber bands
2. Geometry box
3. Two pens or pencils



Figure 1: Step 1. Push the free end of the scale downwards.

Image credit: Ingrid Sulston, licensed under CC BY-NC-SA 4.0



Figure 2: Step 2. Let the pushed end go.

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Close the geometry box or pencil box. Stretch a rubber band around the box so that it lies across the top surface. Place two pens or pencils under the rubber band near the two ends. This will lift the rubber band slightly above the surface of the box. See Figure 3. Pluck the rubber band with your finger.

OBSERVE: Do you hear a sound? What part of the setup is moving?

Optional Activity: Tuning Fork

If a tuning fork is available, try this activity.

Strike a prong of a tuning fork gently with a rubber hammer to make it vibrate. Bring the vibrating tuning fork close to your ear.

OBSERVE: can you hear a sound?

Gently touch the prongs of the vibrating tuning fork with your finger. What do you feel? Strike the tuning fork again. Lightly touch the vibrating prongs to the surface of water in a bowl.

OBSERVE: do you see ripples forming on the surface of the water?

DISCUSSION: In each case, some part of the object moves back and forth rapidly. This type of motion is called **vibration**.

SOUND IS PRODUCED WHEN OBJECTS VIBRATE. The vibrating part of an object can make nearby air move. These vibrations spread through the air and reach our ears as sound.

EXERCISE: create a table as shown in Table listing the object that produces sound and the part of the object that vibrates.

Sl. No.	Object that produces sound	Part of the object that vibrates
1	Vibrating scale	Tip of the scale
2	Musical geometry box	⋮
3	⋮	⋮
⋮	⋮	⋮

Image of a pencil box with a rubber band around it.

Figure 3: Pencil box with rubber band around it.



Figure 4: Tuning fork

Did you know? Many quartz clocks and watches keep time using a small quartz crystal. This crystal is shaped like a tiny tuning fork.

Table 1: Observation table

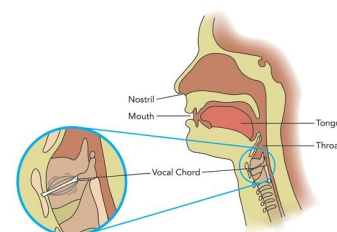


Figure 5: Our vocal chords vibrate when we produce sound. Image source: SASOL Inzalo Foundation

How Does Sound Propagate?

String Telephone

Materials required:

1. Two paper cups
2. A long piece of string

Make a small hole at the bottom of each cup. Tie the ends of the string to the cups through these holes. Give one cup to your benchmate. Stand away from your benchmate to keep the string tight. Speak into one cup while your benchmate listens through the other cup. See Figure ??

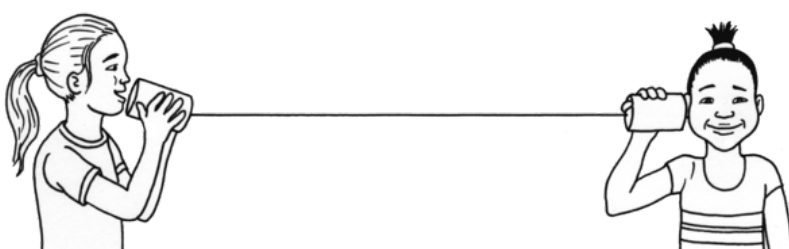


Figure 6: Image source: SASOL Inzalo Foundation

OBSERVE: Can your benchmate hear the voice? What happens to the sound when you let the string loose?

DISCUSSION: When the string is tight, vibrations travel through the string. When the string is loose, the vibrations do not travel well. This shows that sound needs a material medium to propagate.

Activity: Listening Through Wood or Metal

Place your ear on the desk. Ask your benchmate to gently tap the desk with a pencil a short distance away. Now listen again with your ear above the desk in the air.

OBSERVE: Which sound is louder or clearer? The sound through the table or the sound through the air?

DISCUSSION: The sound is usually clearer through the table. This shows that sound can propagate through solids, not only through air.

Sound can propagate through gases, liquids, and solids. A material through which sound propagates is called a **medium**.

Sound cannot propagate in the absence of a medium. In other words, sound cannot travel through a vacuum.

QUESTION: Why can astronauts not hear each other directly on the Moon?

Decibel (dB) is the unit of measurement for sound level.

There is no air on the Moon. If two people stand a short distance apart on the Moon and shout, they cannot hear each other. Because of this, astronauts on the Moon use radios to communicate with each other.

Did you know? Radios use **electromagnetic waves**. Electromagnetic waves can travel through empty space.

Activity: Does air travel with the sound?

Materials required:

1. A piece of tissue paper or a candle
2. A sound-producing object (such as a bell, or a plate and spoons)

Place the candle (or hold the tissue paper) about 20-30 cm away from the sound source. Produce a sound.

OBSERVE: Watch the flame or tissue carefully. Does the tissue or the flame of the candle move? Does it move only slightly, or not at all?

QUESTION: If air were travelling from the sound source to your ear like wind, what should happen to the flame or tissue?

DISCUSSION: In sound, the air does not flow across the room. Instead, the air vibrates back and forth in place. Each part of the air passes the vibration to the next part. In this way, the vibration propagates through the medium and reaches our ears as sound.

QUESTION: When you hear a sound, what is actually moving between the source of sound and your ear?

Sound Waves

We have seen that sound is produced when objects vibrate. These vibrations make the surrounding air vibrate. The air particles move back and forth around their positions. Each particle pushes and pulls its neighbouring particles. In this way, the vibration passes through the air from the sound source to our ears. When vibrations move through a medium like this, they form a wave. Sound propagates through air as a sound wave. See Figure .

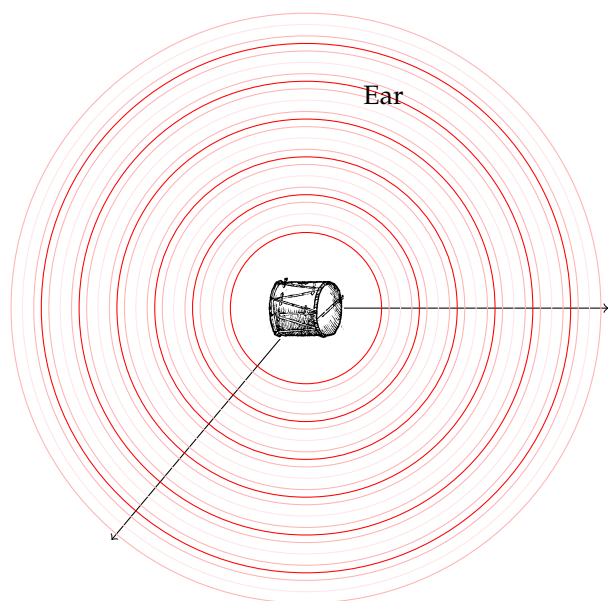


Figure 7: Sound wave propagates from the source of sound (drum in this case) to the listener in the form of vibrations in the medium.

Compressions and Rarefactions

When sound propagates through air, the air particles move back and forth around their positions. As they move, the particles sometimes come closer together and sometimes move farther apart. Regions where the air particles are closely packed are called **compressions**. Regions where the particles are farther apart are called **rarefactions**. See Figure ??.

As a sound wave propagates, compressions and rarefactions move forward through the air. This pattern of compressions and rarefactions carries sound from the source to our ears.

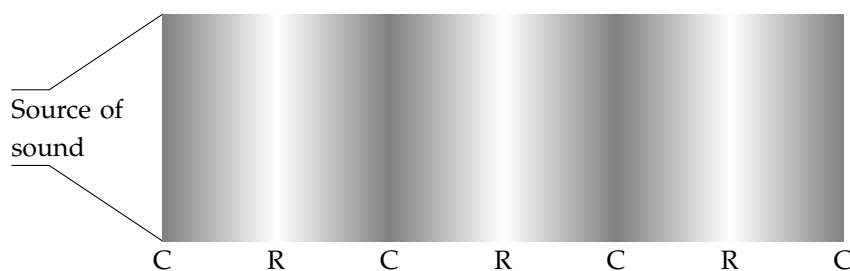


Figure 8: Sound propagates as density variations in the medium. C: Compression, R: Rarefaction

Optional Activity: Long Spring

Materials required:

1. A long spring or slinky
2. A small ribbon

Hold one end of the spring while your classmate holds the other end. Stretch the spring gently between you. Now give the spring a quick push and pull along its length. Tie a ribbon somewhere in the middle of the spring and watch it carefully.

OBSERVE: What kind of motion does the ribbon show?

DISCUSSION: You will notice that the ribbon moves back and forth along the length of the spring. The ribbon does not travel along the spring. Instead, it moves back and forth around its position. In a similar way, air particles move back and forth when sound propagates.

In such waves, the particles of the medium move parallel to the direction in which the wave propagates. Such waves are called **longitudinal waves**. Sound propagates as a longitudinal wave through solids, liquids, and gases.



Figure 9: Long spring also known as a slinky.

Frequency of Sound

Activity: Vibrating Scale

Place a scale on the edge of a desk so that part of it stays over the edge. Hold the scale firmly on the desk and pluck the free end so that it vibrates and produces a sound. Optionally, you may record a slow-motion video using a phone.

OBSERVE: How does the sound change when the length of the overhanging part changes? Does the scale vibrate faster or slower?

You may notice that when the scale is longer, it vibrates more slowly and the sound is lower. When the scale is shorter, it vibrates faster and the sound is higher. If you record a slow-motion video, you can count how many times the scale moves back and forth in 5 seconds. Divide this number by 5 to find how many vibrations occur in one second.

Observation No.	No. of times the scale moves back and forth in 5 seconds (n)	$n/5$
1	...	...
2	...	...
3	...	...

Table 2: Observation table for finding the frequency of the vibrating scale

When the scale vibrates, it also makes the surrounding air vibrate. As the scale moves back and forth, it pushes and pulls the nearby air. This creates compressions and rarefactions in the air.

DISCUSSION: the number of vibrations that occur in one second is called the **frequency** of the vibration. Frequency is measured in hertz (Hz) and usually represented by the Greek letter ν (nu). Frequency tells us how often an event occurs in a unit of time.

EXAMPLE: if you beat a drum 10 times in one second, the frequency of your beating is 10 Hz.

Pitch

Sounds with higher frequency have a higher pitch and sounds with lower frequency have a lower pitch. A mosquito makes a high-pitched sound. A drum makes a low-pitched sound. Which one is vibrating faster?

QUESTION: Which one is vibrating faster? Mosquito's wings or the membrane of the drum?

QUESTION: A bell is struck and produces sound. After a few seconds, the sound stops. What must have happened to the bell?

QUESTION: Kavin and Priya are standing at different distances from the lightning. Both hear the thunder. Will the pitch of the thunder be different for them? Will the loudness of the thunder be different?

Amplitude

You may also have noticed something else during the activity. When the scale was plucked gently, the sound was soft. When the scale was plucked harder, the sound was louder.

If you watch the scale carefully, you will see that when it is plucked harder, it moves farther away from its resting position during each vibration.

The size of this vibration is called the **amplitude**. Amplitude tells us how large the vibration is. A vibration with greater amplitude produces a louder sound, while a vibration with smaller amplitude produces a softer sound. Amplitude affects loudness, not pitch.

QUESTION: Two sounds have the same pitch, but one is louder. What must be different?

Hearing loss and hearing aids

Very loud sounds can damage the inner ear and lead to hearing loss. Some people are born with conditions that affect their ability to hear.

Hearing aids are devices that help such people hear better. They work by amplifying sound, that is, by increasing the amplitude of the sound waves.

Activity: long spring

Take a long spring. Ask your classmate to hold one end while you hold the other end. Stretch the long spring gently between you. Now give the long spring a quick push toward your classmate.

OBSERVE: You will see regions where the coils are close together and regions where they are farther apart. A region where the coils are close together is called a compression. A region where the coils are farther apart is called a rarefaction. This is similar to what happens in sound waves in air.

Wavelength

The distance between two neighbouring compressions (or two neighbouring rarefactions) is called the **wavelength**. Wavelength is measured in the units of distance. The SI unit of distance is the metre (m). Wavelength is usually represented by the Greek letter λ (lambda). See Figure . Repeat the activity, but now create compressions more quickly.

Pitch depends on frequency, which is determined by the source. Distance from the source does not change the frequency. As sound spreads out, its amplitude decreases, this is why it becomes quieter as you move far from the source.

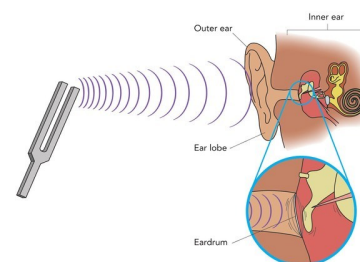


Figure 10: Diagram of the auditory system of the human ear. Image source: SASOL Inzalo Foundation.

Image of hearing aid

Figure 11: Image of hearing aid

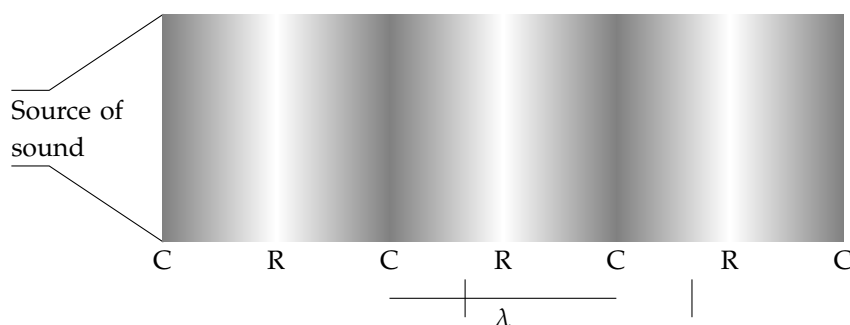


Figure 12: The distance between two neighbouring compressions (or two neighbouring rarefactions) is the wavelength (λ).

OBSERVE: When vibrations happen more often each second (higher frequency), the compressions are packed closer together and the wavelength becomes shorter.

Speed of Sound

Do we hear sound instantly after it is produced? Think about what happens during lightning and thunder. When lightning strikes, we see a flash of light and hear thunder.

OBSERVE: Do you see the lightning and hear the thunder at the same time? Or do they happen one after the other? Which do you notice first?

Usually, we see the lightning first and hear the thunder later.

These observations show that sound takes time to propagate from the source to our ears. Light travels much faster than sound, so we see the flash before we hear the sound.

In air, sound propagates at a speed of about 340 metres per second (at room temperature). The speed of a sound wave depends on both its frequency and its wavelength.

$$\text{Speed} = \text{Frequency} \times \text{Wavelength} \quad (1)$$

Sound does not propagate at the same speed in all materials.

The speed of sound depends on the medium through which it propagates. Sound propagates fastest in solids, slower in liquids, and slowest in gases. For example, sound propagates faster in metal or wood than in air. This is because the particles in solids are closer together, allowing vibrations to pass more quickly from one particle to the next.

QUESTION: How does temperature affect the speed of sound?

Musical instruments

Musical instruments produce sound by making strings, membranes, or air columns vibrate. By changing certain parts of the instrument, musicians can change the frequency of vibration, and therefore the pitch of the sound.



Figure 13: Thunderstorm in Tiruchirappalli, Photo by Amol Mande
If the thunder sounds quieter when you are farther away, does the thunder have less energy when it reaches you?

As the temperature increases, the speed of sound increases. This happens because when the temperature is higher, the particles of the medium move faster. As a result, they pass the vibrations to neighbouring particles more quickly, allowing sound to propagate faster. For example, sound travels faster in warm air than in cold air.

Different instruments change the frequency in different ways.

Veena (string instrument): The strings vibrate to produce sound. Pressing the string at different positions changes the length of the vibrating string, which changes the frequency and the pitch.

Parai (drum): The stretched membrane (skin) vibrates when it is struck. Changing the tightness of the membrane changes the frequency of vibration and the sound produced.

Nadaswaram (wind instrument): Sound is produced by vibrations of the air column inside the instrument. Opening and closing the finger holes changes the effective length of the air column, which changes the frequency.

Although these instruments look very different, they all produce sound through vibrations, and the frequency of vibration determines the pitch of the sound.

Ultrasound

Humans can hear sounds with frequencies roughly between 20 Hz and 20,000 Hz. Sounds with frequencies higher than 20,000 Hz are called ultrasound. Humans cannot hear ultrasound.

Ultrasound has many useful applications:

1. **Medical imaging:** Doctors use ultrasound to look inside the body, for example to observe a developing fetus during pregnancy.
2. **Detecting cracks in materials:** Ultrasound can be used to find cracks or defects in metals and other materials.
3. **Animal communication and navigation:** Some animals, such as bats and dolphins, produce and detect ultrasound. They use it to communicate and to find objects around them.

Echo

Sound travels at a finite speed. When a sound wave meets a surface such as a wall, cliff, or building, it can reflect and travel back toward the source.

An echo is heard when the reflected sound reaches our ears after a short delay.

For an echo to be heard clearly, the reflecting surface must be far enough away. If the surface is too close, the reflected sound mixes with the original sound and we do not hear a separate echo.

For example, when you shout in a large empty hall, valley, or near a cliff, you may hear your own voice repeated after a short time. This repeated sound is the echo. Echoes show that sound waves can reflect from surfaces and travel back to us.

Echolocation

Animals such as bats and dolphins produce sounds and listen to the echoes to locate objects around them.



Figure 14: Veena from Chennai, Wikimedia Commons



Figure 15: Front side of a parai with the sticks used to play the instrument. Image by Prabhunkl, licensed under CC BY-SA 4.0



Figure 16: Nadaswaram (also nagaswaram or nagasuram), Image by Ashwin Kamath

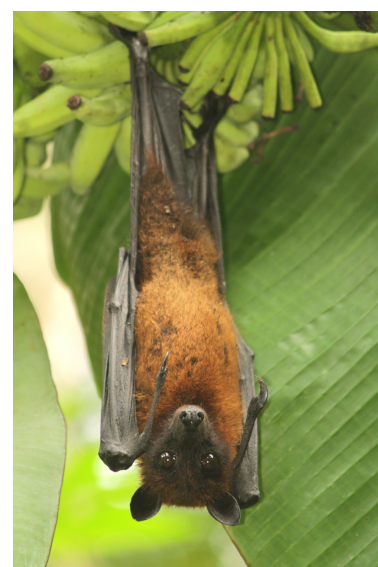


Figure 17: Greater Indian fruit bat, image by Praveenp, licensed under CC BY-SA 4.0

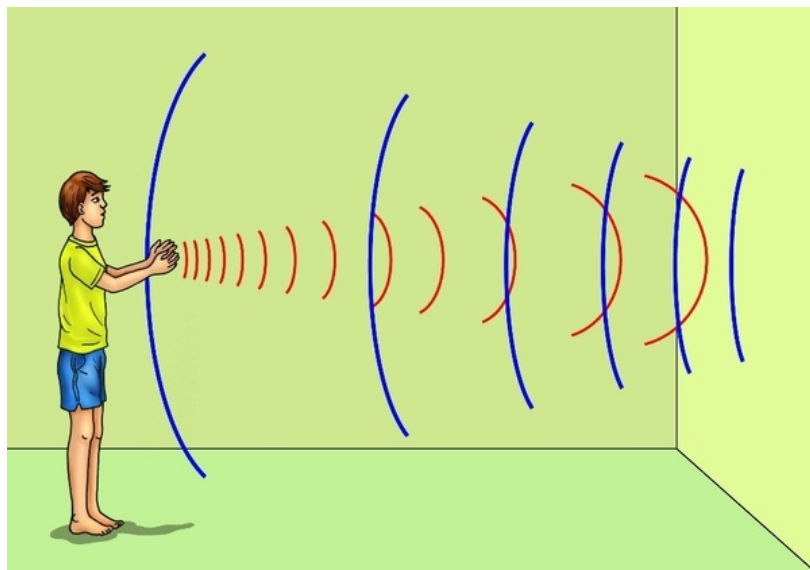


Figure 18: Image source: SASOL Inzalo Foundation

Sonar

Ships and submarines use sound waves and their echoes to detect objects underwater and measure distances.

Noise Pollution

Unwanted or unpleasant sound is called noise. When there is too much noise in our surroundings, it leads to noise pollution.

Common sources of noise pollution include traffic noise from vehicles and horns, loudspeakers used at high volume, industrial noise from machines and factories, construction activities.

Continuous exposure to loud noise can cause stress, lack of sleep, and hearing problems. How can we reduce noise pollution?

We can reduce noise pollution by avoiding unnecessary honking on the roads, keeping the volume of loudspeakers low, using silencers and maintaining vehicles properly, planting trees, which can help absorb sound, following rules and regulations about noise levels.

Reducing noise pollution helps create a healthier and more peaceful environment.

Sound is an example of a mechanical wave. It is also a longitudinal wave, because the particles of the medium vibrate along the direction in which the wave propagates. There is another type of wave called a transverse wave. In a transverse wave, the particles of the medium move perpendicular to the direction in which the wave propagates. In other words, the particles move up and down, while the wave moves forward.

An example of a transverse wave can be seen when a pebble is

If you pick up a seashell on a beach and hold it close to your ear, you may hear a sound that resembles waves breaking on the shore. Where does this sound come from?

dropped into a pond. The ripples that spread across the surface of the water are transverse waves.

Light is also a transverse wave. However, light does not need a material medium to propagate, so it is not a mechanical wave.

The world around us is full of waves - sound waves, water waves, and light waves.

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